Parth A. Tokekar

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Education

International Institute of Information Technology, Hyderabad (IIIT-H) **Aug'23 – May'27**
B.Tech in Electronics and Communication Engineering (ECE) **CGPA: 8.58/10**
School of Scholars, Nagpur **2021 – 2023**
Class XI – XII (CBSE) **Percentage: 96%**

Experience

SDE Intern, Salk AI **May'25 – July'25**
Meta Graph API, OpenAI Agents SDK, LiveKit

- Engineered a production-grade voice interaction platform using **LiveKit**, enabling intelligent call workflows that processed **30+ inbound leads/day**, with real-time voice intent extraction and automated lead triaging.
- Designed and deployed agentic pipelines for WhatsApp engagement, leveraging **OpenAI Agents SDK** to auto-tag and rank **250+ daily conversations**, improving high-quality lead routing by **45%**.
- Integrated WhatsApp and CRM systems to deliver **contextually aware, event-triggered follow-ups**, reducing manual operational effort by **60%** and cutting lead response latency by **40%**.

Undergraduate Researcher, SPCRC, IIIT Hyderabad **Apr'25 – Present**
Signal Processing and Communication Research Center (SPCRC)

- Researching signal estimation techniques for wireless receivers, focused on inferring latent transmission parameters (e.g., channel coefficients, SNR, Doppler shift) under unknown conditions.
- Developing hybrid inference pipelines that combine **classical statistical signal processing** (e.g., Kalman filtering, matched filtering) with supervised **ML models** (SVMs, regression nets) to improve robustness under noisy channel assumptions.

SDE Intern, Salk AI **June'24 – July'24**
LangChain, Hugging Face, OpenAI API

- Architected modular **Retrieval-Augmented Generation (RAG)** pipelines for property-tech clients, integrating **LangChain**, **LLM APIs**, and **HuggingFace** models for domain-specific semantic search.
- Reduced retrieval latency by **35%** and increased response throughput by **3×** through asynchronous request handling and index-level optimizations.

Projects

AI-Powered Cricket Reels Generator *Gemini 2.5, Whisper*
FastAPI, ffmpeg, HeyGen

- Built a **FastAPI** backend to generate cricket highlight reels from daily facts in under **5 minutes**.
- Used **Gemini 2.5** for scriptwriting and **Whisper** for transcription with timestamps to enable precise video overlays.
- Automated avatar video creation using **HeyGen API**, reducing manual editing time by **90%**.

High-Performance Processor Design *Verilog, RISC-V, Pipelining*
Computer Architecture, ISA Design

- Developed a 5-stage pipelined **RISC-V** processor with instruction fetch, decode, execute, memory, and write-back units.
- Implemented hazard detection and data forwarding, reducing stalls and improving latency by **20%** and CPI by **10%**.
- Created testbenches and simulated edge cases to validate correctness and performance.

Technical Skills

Languages: Python, C/C++, Bash, SQL, Assembly, MATLAB, Verilog
Frameworks: FastAPI, LangChain, Hugging Face, Flask, PyTorch, NumPy, Pandas
Tools & Platforms: Git, Docker, Google Cloud Platform, Postgres, MongoDB, Linear, PgAdmin
Concepts: REST APIs, RAG Systems, Signal Processing, NLP, LLMOps, Pipelining, CRM Integration

Achievements

- Secured an **All India Rank (AIR) of 2609** in JEE Mains (2023), among top 0.5% of candidates nationwide.
- Achieved an **AIR of 3888** in JEE Advanced (2023), among 2lakh+ candidates.
- Awarded the **Academic Excellence Award** for being in the top 20% of the batch at IIIT Hyderabad (2024).

Leadership & Activities

Corporate Head, Felicity'26 (College Fest) *Mar 2025 – Present*
Coordinator, Comedy Club *Jul 2024 – Present*